

Iterator Granularity Sweep

Overfit scaling vs encoding resolution · Cross-domain

Train: Genesis Ch.1–10 · Test: Alice Ch.I

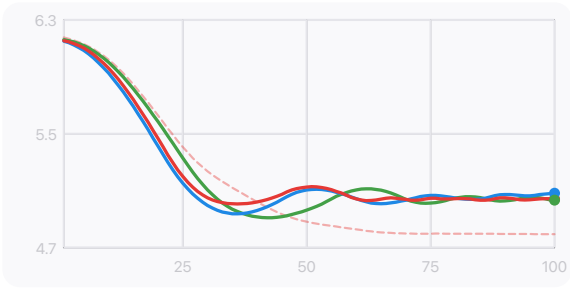
D=3 fixed input CTX=32 128×2 MLP 100 epochs lr=0.003 Vw=482 seed=42

LOSS CURVES · TRAIN (DASHED) VS TEST (SOLID)

Iterator (C) test - - - Iterator train Raw (A) test Random (R) test

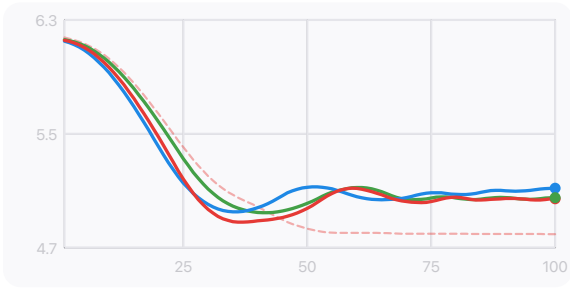
W=1

1 bit per token · 1-bit



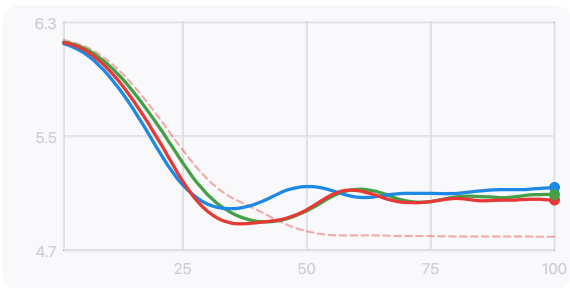
W=8

8 bits per token · character



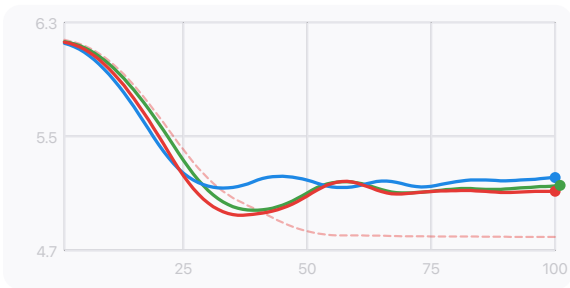
W=16

16 bits per token · bigram



W=24

24 bits per token · triplet



GENERALISATION GAP (TEST – TRAIN) AT EPOCH 100

RESOLUTION	C GAP	R GAP	A GAP	C – R
W=1 1-bit	+0.2484	+0.2389	+0.5408	+0.0095
W=8 character	+0.2497	+0.2739	+0.5434	-0.0242

W=16 bigram	+0.2590	+0.3327	+0.5361	-0.0737
W=24 triplet	+0.3270	+0.4258	+0.6341	-0.0988

KEY FINDING

C is granularity-stable

Iterator overfit range: +0.248 to +0.327 across 24x resolution change

C - R TREND

Monotone separation

Iterator advantage over random accumulates continuously with window size

CONDITIONS

LABEL	DESCRIPTION
C · Iterator	[mean_b0, mean_b1, D*_digest] · quad averaged over W bits · digest at word boundary · fixed input
A · Raw	Raw triplet tokens · learned D=3 embedding · trained end-to-end
R · Random	Gaussian noise D=3 · fixed input · control condition